Q1

May 1, 2025

1 Q1 (Causal Inference): What is the causal effect of high predicted PM2.5 concentrations on adult asthma prevalence in California during 2018?

[2]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
from sklearn.linear_model import LinearRegression
import statsmodels.api as sm
from sklearn.mixture import GaussianMixture
from sklearn.linear_model import LogisticRegression

sns.set(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
```

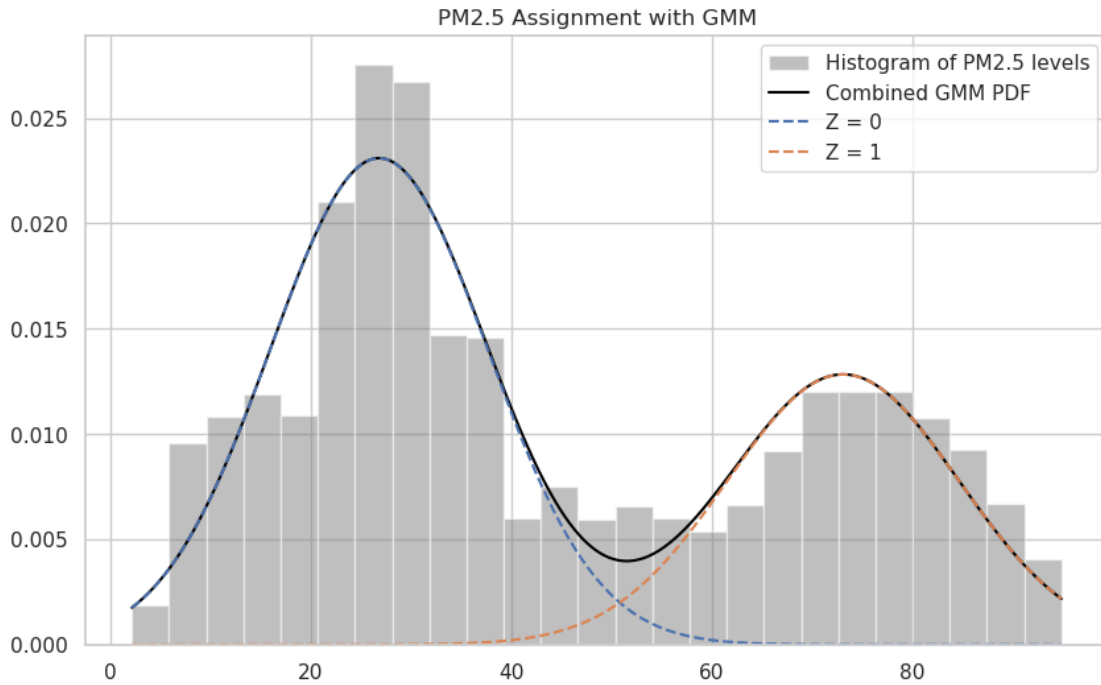
[3]:
```
df = pd.read_csv("data_merged_cleaned.csv")

print(df.shape)
df.head()
```

```
(8005, 60)
```

[3]:
```
   Year StateAbbr   StateDesc CountyName  CountyFIPS  LocationName DataSource  \
0  2018        CA  California    Alameda        6001    6001400100      BRFSS
1  2018        CA  California    Alameda        6001    6001400200      BRFSS
2  2018        CA  California    Alameda        6001    6001400300      BRFSS
3  2018        CA  California    Alameda        6001    6001400400      BRFSS
4  2018        CA  California    Alameda        6001    6001400500      BRFSS

          Category                                       Measure  \
0  Health Outcomes  Current asthma among adults aged >=18 years
1  Health Outcomes  Current asthma among adults aged >=18 years
2  Health Outcomes  Current asthma among adults aged >=18 years
3  Health Outcomes  Current asthma among adults aged >=18 years
4  Health Outcomes  Current asthma among adults aged >=18 years
```

```
  Data_Value_Unit  … Walk  OtherTransp  WorkAtHome  MeanCommute  Employed  \
0               %  …  0.5          2.2        20.0         34.8    1543.0
1               %  …  4.4          5.0        11.8         29.2    1231.0
2               %  …  5.4          4.9        10.3         31.3    3213.0
3               %  …  4.3         12.8         9.1         30.4    2499.0
4               %  …  2.9         11.1        14.0         32.4    2587.0

   PrivateWork  PublicWork SelfEmployed  FamilyWork Unemployment
0         80.6         6.6         12.8         0.0          4.0
1         77.4        10.2         11.6         0.8          3.7
2         73.0        17.8          9.2         0.0          5.5
3         73.9        15.2         10.7         0.3          5.0
4         77.1        14.1          8.8         0.0          2.6

[5 rows x 60 columns]
```

[4]:
```
pm_gmm = df['DS_PM_pred'].to_numpy().reshape(-1, 1)

gmm = GaussianMixture(n_components=2, random_state=0)
gmm.fit(pm_gmm)

counts, bins, _ = plt.hist(pm_gmm, bins=25, density=True, alpha=0.5,␣
 ↪color="gray", label="Histogram of PM2.5 levels")

x = np.linspace(pm_gmm.min(), pm_gmm.max(), 1000).reshape(-1, 1)
log_prob = gmm.score_samples(x)
pdf = np.exp(log_prob)
plt.plot(x, pdf, label="Combined GMM PDF", color="black")

for i in range(2):
    mean = gmm.means_[i][0]
    std = np.sqrt(gmm.covariances_[i][0][0])
    weight = gmm.weights_[i]
    component_pdf = weight * (1 / (std * np.sqrt(2 * np.pi))) * np.exp(-0.5 *␣
 ↪((x - mean) / std) ** 2)
    plt.plot(x, component_pdf, "--", label=f"Z = {i}")

plt.legend()
plt.title("PM2.5 Assignment with GMM")
plt.show()
```

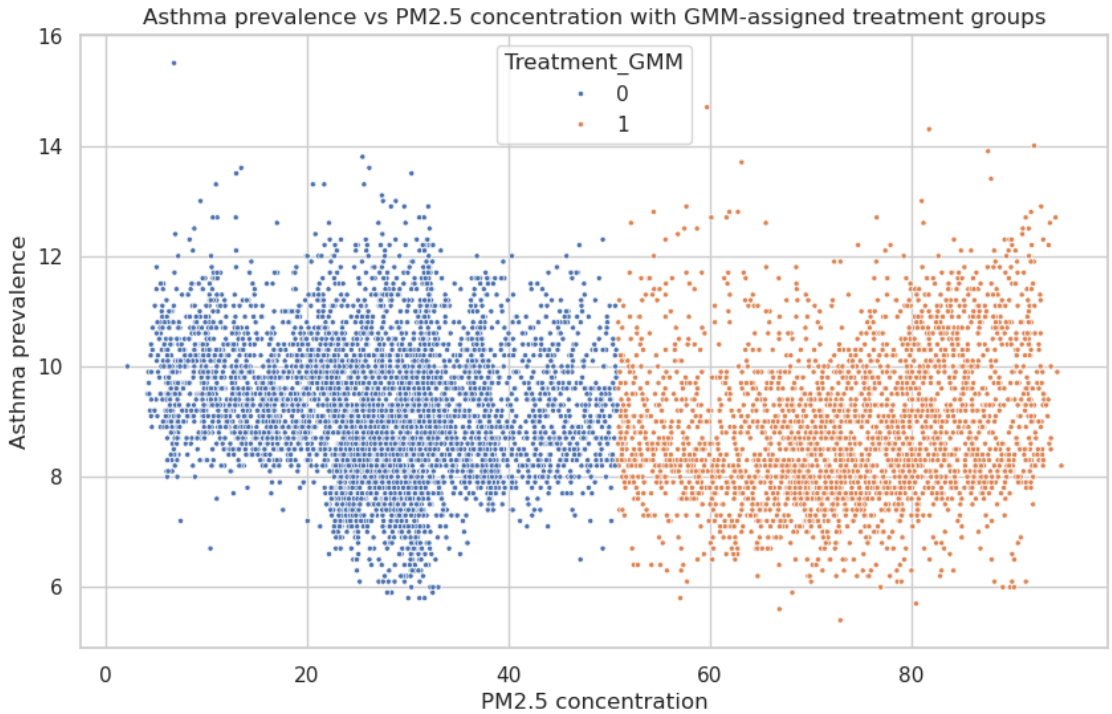


[5]:
```
responsibilities = gmm.predict_proba(pm_gmm)
assignments = np.argmax(responsibilities, axis=1)
df["Treatment_GMM"] = assignments
```

Used a Gaussian Mixture Model (GMM) to fit two distributions to the bimodal distribution of PM2.5 concentrations, then used those two distributions to assign data points to treatment groups. Each data point was assigned to the group whose distribution had a higher PDF at that point. The distribution with the lowest mean (left peak in the graph above) corresponds to low PM2.5 concentrations, and therefore the group where treatment equals 0 since the treatment is being in a high PM2.5 area.

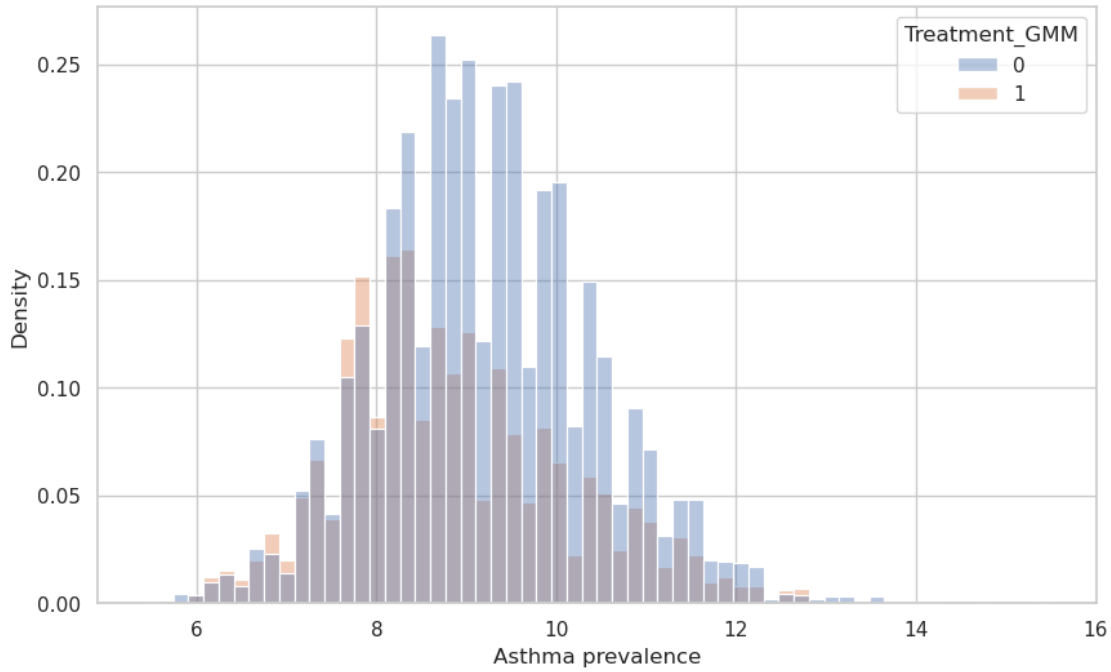
[6]:
```
sns.scatterplot(data=df, x="DS_PM_pred", y="Data_Value", marker=".",␣
↪hue="Treatment_GMM")
plt.xlabel("PM2.5 concentration")
plt.ylabel("Asthma prevalence")
plt.title("Asthma prevalence vs PM2.5 concentration with GMM-assigned treatment␣
↪groups")
```

[6]:
```
Text(0.5, 1.0, 'Asthma prevalence vs PM2.5 concentration with GMM-assigned
treatment groups')
```

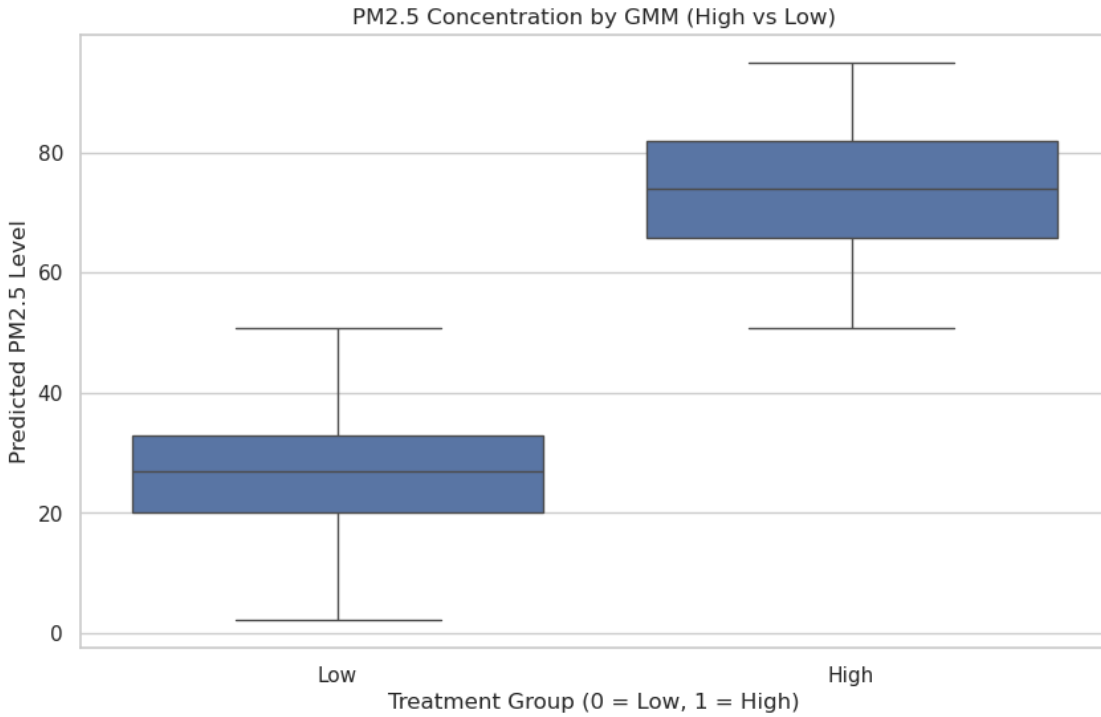


[7]:
```
sns.histplot(data=df, stat="density", x="Data_Value", hue="Treatment_GMM",␣
↪alpha=0.4)
plt.xlabel("Asthma prevalence")
```

[7]: Text(0.5, 0, 'Asthma prevalence')



[8]:
```
sns.boxplot(x='Treatment_GMM', y='DS_PM_pred', data=df)

plt.title('PM2.5 Concentration by GMM (High vs Low)')
plt.xlabel('Treatment Group (0 = Low, 1 = High)')
plt.ylabel('Predicted PM2.5 Level')
plt.xticks([0, 1], ['Low', 'High'])

plt.show()
```



Observations:

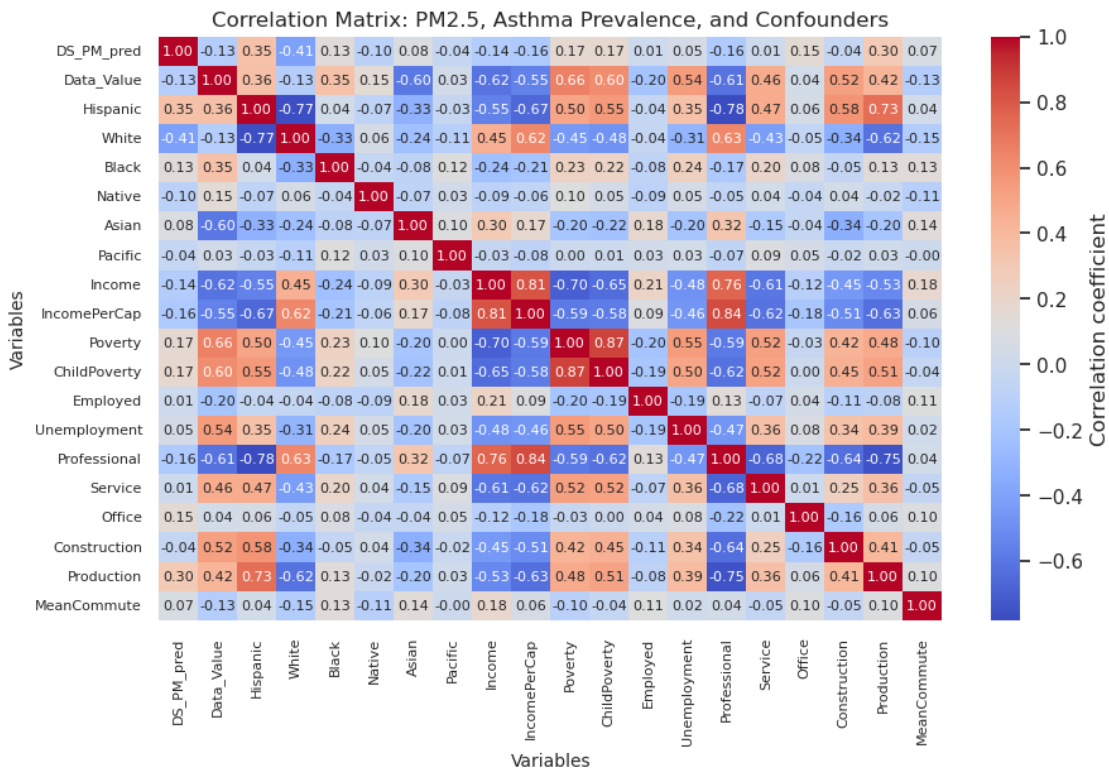
The plot clearly confirms that your PM2.5 clustering (into "High" and "Low") is meaningful and distinct. The median PM2.5 level in the high cluster is substantially higher (around 75) than in the low cluster (around 28). There is little overlap in interquartile ranges (IQRs), which suggests strong separation between the two groups.

Relevance to Question:

This clustering provides a binary treatment variable (High vs. Low PM2.5) necessary for methods like Inverse Propensity Weighting or matching. It validates that the treatment is not arbitrary—we're distinguishing areas that truly differ in pollution exposure. This setup helps us estimate the causal effect of high PM2.5 on asthma while adjusting for confounders like race or gender.

[9]:
```
confounders = ['Hispanic', 'White', 'Black', 'Native', 'Asian', 'Pacific',␣
↪'Income', 'IncomePerCap',
               'Poverty', 'ChildPoverty', 'Employed', 'Unemployment',␣
↪'Professional',
               'Service', 'Office', 'Construction', 'Production', 'MeanCommute']

correlations = df[['DS_PM_pred', 'Data_Value'] + confounders].corr()
sns.heatmap(correlations, annot=True, cmap='coolwarm', fmt=".2f",␣
↪annot_kws={"size": 8},
            cbar_kws={'label': 'Correlation coefficient'})
```

```
plt.title("Correlation Matrix: PM2.5, Asthma Prevalence, and Confounders",␣
↪fontsize=12)
plt.xlabel('Variables', fontsize=10)
plt.ylabel('Variables', fontsize=10)
plt.xticks(fontsize=8)
plt.yticks(fontsize=8)
plt.show()
```



Observations:

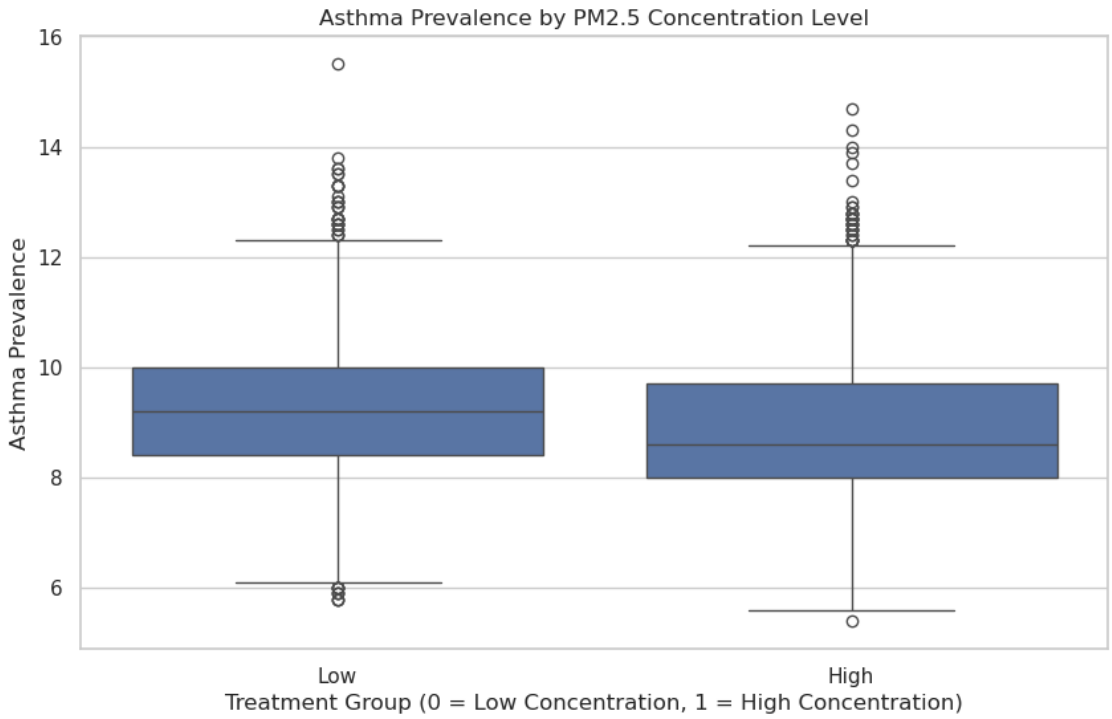
The first potential confounder is production, which are correlated with both income and asthma, which may influence modeling assumptions. Poverty and income are highly inversely correlated (-0.70), which is expected.

PM2.5 has negative correlation with income (-0.14), professional employment (-0.16). Positive correlation with poverty (0.17) and production sector employment (0.30). Suggests that lower-income, industrial communities may have higher pollution levels.

Relevance to Question:

The matrix supports the presence of important confounders: prodcution, income, poverty, and race (Hispanic) are related both to PM2.5 exposure and asthma prevalence. These need to be carefully controlled for or stratified in causal models like IPW or matching.

[10]:
```
sns.boxplot(x='Treatment_GMM', y='Data_Value', data=df)

plt.title('Asthma Prevalence by PM2.5 Concentration Level')
plt.xlabel('Treatment Group (0 = Low Concentration, 1 = High Concentration)')
plt.ylabel('Asthma Prevalence')
plt.xticks([0, 1], ['Low', 'High'])

plt.show()
```



[11]:
```
# Drop rows where any confounder is missing
confounders = [
    "Income",
    "IncomePerCap",
    "Poverty",
    "ChildPoverty",
    "Professional",
    "Production",
    "Unemployment",
    "Hispanic",
    "Black",
    "Asian",
    "White",
    "Native",
    "MeanCommute"
```

```
]
df = df.dropna(subset=confounders)

# Create treatment variable: High_PM25 based on 75th percentile of DS_PM_pred
# q3_threshold = df["DS_PM_pred"].quantile(0.75)
# df["High_PM25"] = (df["DS_PM_pred"] > q3_threshold).astype(int)

# Set up X and y for propensity model
X = df[confounders]
y = df["Treatment_GMM"]

# Fit logistic regression to estimate propensity scores
ps_model = LogisticRegression(max_iter=1000)
ps_model.fit(X, y)

# Predict propensity scores
df["propensity_score"] = ps_model.predict_proba(X)[:, 1]

# Preview results
df[["Treatment_GMM", "propensity_score"]].head()
```

```
/srv/conda/lib/python3.11/site-packages/sklearn/linear_model/_logistic.py:465:
ConvergenceWarning: lbfgs failed to converge (status=1):
STOP: TOTAL NO. of ITERATIONS REACHED LIMIT.

Increase the number of iterations (max_iter) or scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-
regression
  n_iter_i = _check_optimize_result(
```

[11]:

	Treatment_GMM	propensity_score
0	0	0.216633
1	0	0.331652
2	0	0.455777
3	0	0.483954
4	0	0.344810

[12]:

```
df["ipw_weight"] = df["Treatment_GMM"] / df["propensity_score"] + (1 -␣
↪df["Treatment_GMM"]) / (1 - df["propensity_score"])

df[["Treatment_GMM", "propensity_score", "ipw_weight"]].head()
```

[12]:

	Treatment_GMM	propensity_score	ipw_weight
0	0	0.216633	1.276541
1	0	0.331652	1.496227

```
2                0          0.455777    1.837483
3                0          0.483954    1.937810
4                0          0.344810    1.526276
```

Applied inverse probability weighting (IPW) to adjust for confounding. Each census tract was weighted inversely proportional to its probability of receiving the treatment it actually received.

[13]:
```
treated_mean = (df[df["Treatment_GMM"] == 1]["Data_Value"] *␣
↪df[df["Treatment_GMM"] == 1]["ipw_weight"]).sum() / df[df["Treatment_GMM"]␣
↪== 1]["ipw_weight"].sum()

control_mean = (df[df["Treatment_GMM"] == 0]["Data_Value"] *␣
↪df[df["Treatment_GMM"] == 0]["ipw_weight"]).sum() / df[df["Treatment_GMM"]␣
↪== 0]["ipw_weight"].sum()

ate = treated_mean - control_mean

print(f"Weighted mean Asthma Prevalence (High PM2.5): {treated_mean:.4f}")
print(f"Weighted mean Asthma Prevalence (Low PM2.5): {control_mean:.4f}")
print(f"Estimated Average Treatment Effect (ATE): {ate:.4f}")
```

```
Weighted mean Asthma Prevalence (High PM2.5): 8.7987
Weighted mean Asthma Prevalence (Low PM2.5): 9.1725
Estimated Average Treatment Effect (ATE): -0.3738
```

Used inverse probability weighting to estimate the causal effect of high PM2.5 exposure on asthma prevalence. The ATE represents the estimated difference in average asthma prevalence between high and low exposure census tracts after adjusting for confounding.

In our causal analysis, we investigated the effect of high PM2.5 exposure on adult asthma prevalence in California census tracts for the year 2018. We defined high exposure based on unsupervised clustering (using KMeans or Gaussian Mixture Models) of PM2.5 concentrations to seperate tracts into high and low exposure groups. To adjust for confounding, we estimated propensity scores using logistic regression based on socioeconomic and demographic variables, including income, poverty, race/ethnicity, employment type, and mean commute time. We then applied inverse probability weighting (IPW) to simulate a randomized expirement. After weighting, we found that the weighted average asthma prevalence in high PM2.5 tracts was 8.80%, while in low PM2.5 tracts it was 9.17%. The estimated averate treatment effect(ATE) was -0.38 percentage points.

The ATE was negative which sugests that tracts identified of having higher PM2.5 concentrations actually had slightly lower asthma prevalence after adjusting for confounders. This result is unexpected, given prior research typically links higher air pollution to increased asthma rates. Possible explanations include unmeasured confounding inaccuracies in PM2.5 predictions, or limitations in the clusting method for treatment assignment. Future analyses could explore alternative causal modeling approaches or sensitivity analyses to assess the robustness of this finding.

2 Methods

Treatment: Being in a location with a high PM2.5 concentration. A Gaussian Mixture Model was used to fit two normal distributions to the bimodal distribution of PM2.5 concentrations, then used the two new distributions to assign each data point to the low or high concentration groups.

Outcome: Asthma prevalence.

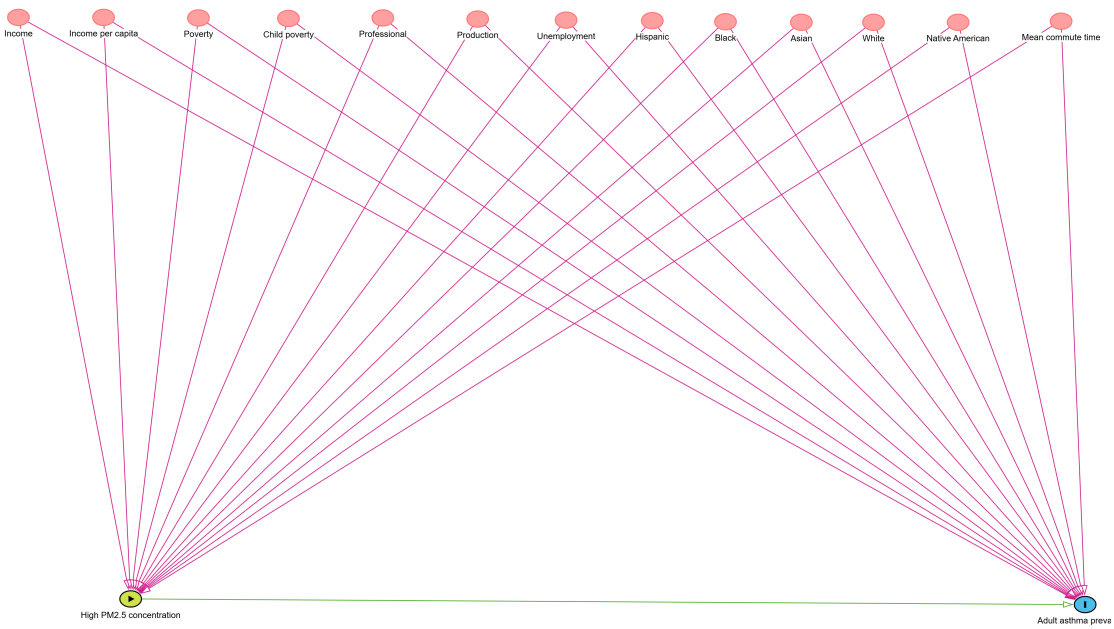
Confounders: Income, income per capita, poverty levels, child poverty levels, race demographics, unemployment levels, employment type, and mean commute time. Inverse Propensity Weighting (IPW) will be used to adjust for these confounders.

Colliders: None

Causal DAG: See image below

[16]:
```
from IPython.display import Image, display
Image("dag.png")
```

[16]:



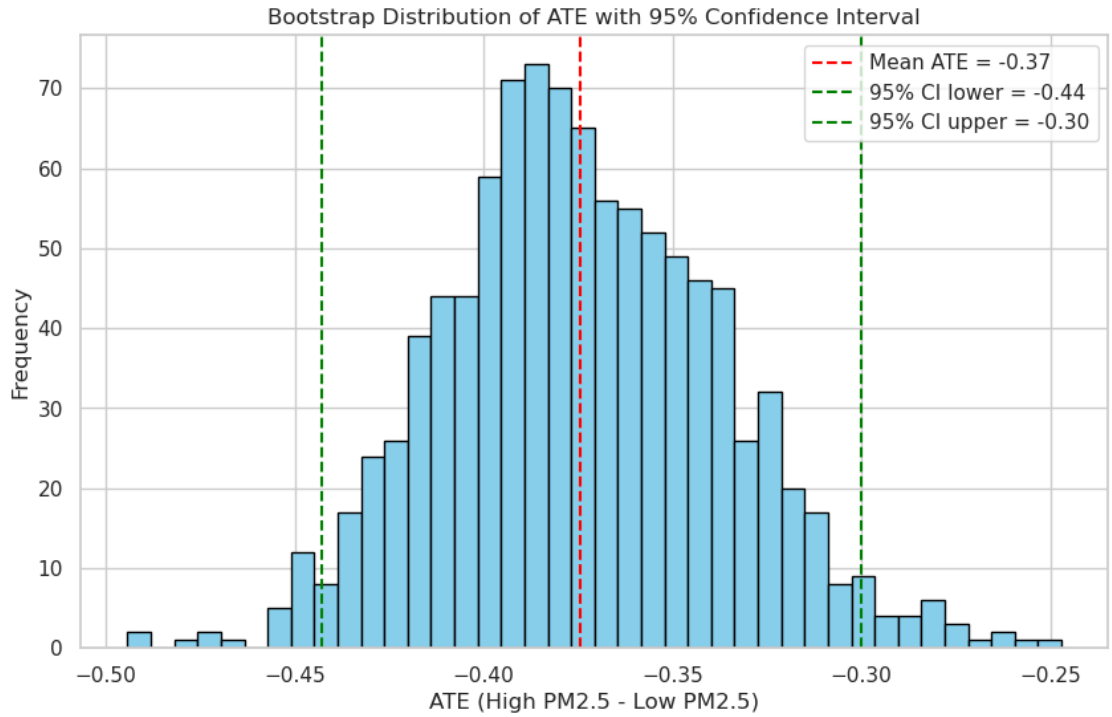
3 Results

After performing inverse propensity weighting to account for confounders, the estimated treatment effect of high PM2.5 concentration on adult asthma prevalence was found to be -0.3761. This indicates that a high PM2.5 concentration in an area decreases its asthma prevalence by about 0.38%.

However, the statistical significance of this small value might be overshadowed by the uncertainties in this experiment, such as any that might be introduced through biases in the data. It is therefore

not clear that there is any causal relationship between PM2.5 levels and asthma prevalence.

[15]:
```
df_boot = df.dropna(subset=["Data_Value", "ipw_weight", "Treatment_GMM"])

n_bootstraps = 1000
bootstrap_ates = []

for _ in range(n_bootstraps):
    sample = df_boot.sample(frac=1, replace=True)
    treated = sample[sample["Treatment_GMM"] == 1]
    control = sample[sample["Treatment_GMM"] == 0]

    treated_mean = (treated["Data_Value"] * treated["ipw_weight"]).sum() /␣
 ↪treated["ipw_weight"].sum()
    control_mean = (control["Data_Value"] * control["ipw_weight"]).sum() /␣
 ↪control["ipw_weight"].sum()

    bootstrap_ates.append(treated_mean - control_mean)

ci_lower = np.percentile(bootstrap_ates, 2.5)
ci_upper = np.percentile(bootstrap_ates, 97.5)
ate_point_estimate = np.mean(bootstrap_ates)

plt.figure(figsize=(10, 6))
plt.hist(bootstrap_ates, bins=40, color="skyblue", edgecolor="black")
plt.axvline(ate_point_estimate, color="red", linestyle="--", label=f"Mean ATE =␣
 ↪{ate_point_estimate:.2f}")
plt.axvline(ci_lower, color="green", linestyle="--", label=f"95% CI lower =␣
 ↪{ci_lower:.2f}")
plt.axvline(ci_upper, color="green", linestyle="--", label=f"95% CI upper =␣
 ↪{ci_upper:.2f}")
plt.title("Bootstrap Distribution of ATE with 95% Confidence Interval")
plt.xlabel("ATE (High PM2.5 - Low PM2.5)")
plt.ylabel("Frequency")
plt.legend()
plt.grid(True)
plt.show()
```



To quanitfy undcertainty in our causal estimate, we applied a bootstrap procedure using 1,000 samples. For each sample, we recalculated the inverse provavility weighted ATE of high PM2.5 exposure on asthma prevalence. The resulting distribution of ATE's is shown in the histogram above. From this, we calculated a 95% confidence interval ranging from -0.45 to -0.30 percentage points, with mean ATE of -0.37. This interval provides evidence that the estimated effect is statistically distinguishable from zero, even though its direction is counterintuitive. The bootstrapping adds robustness to our causal analysis by showing our result is consitent across repeated sampling, and helps communicated the degree of uncertainty surrounding our estimate.

After reviewing the DAG to check for colliders all adjustment variables, including income, race/ethnicity, and employment characteristics, are pretreatment confounders rather than colidders. No collider variables are included in propesnity score model which helps ensure the validity of our causal effect estimation.